

Per-Session KV Store — 3D Memory Architecture

State is replicated across 28 layers, organized per block, and pool-bounded; exact residuals dominate each block's byte budget at the default setting

■ Low-rank core U, V_K, V_V , anchors, min/max (≈ 26 KiB) ■ Exact residuals R_K, R_V (≈ 128 KiB, $R = 128$) ■ Dense recency window (uncompressed fp16)

Compressed block pool · 256 blocks (bounded)

**28-layer stack
($\times 28$ — store
replicated per layer)**

